# Delegating Decisions to AI and Macroeconomic Stability*

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October 2026

**Abstract**

AI systems are beginning to make economic decisions that managers and households used to make. I study the aggregate consequences when firms delegate decisions to a common AI that is better informed than any individual decision-maker. In a linear-quadratic economy with strategic complementarity, delegation replaces dispersed private information with a common signal. Full delegation makes rational expectations literal: the economy aggregates like a representative agent, which removes the attenuation created by dispersed information and raises the response of aggregate output to both fundamentals and the AI's errors. If the AI's precision is fixed, the private and social values of delegation coincide, so higher volatility is not by itself a sign of inefficiency. If the AI learns from the private judgment it replaces, delegation creates a Grossman–Stiglitz externality and is excessive, and complementarity can lock the economy into full delegation to an AI that is less informed than any individual. The policy implication is to protect a floor of independent judgment.

## 1 Introduction

AI systems are beginning to forecast demand, set prices and allocate capital on behalf of firms and households, and each may do so better than the person it replaces. I ask whether better individual decisions make the economy more stable.

Macroeconomics has long modeled agents as if they formed rational expectations while knowing that people do not. The defense has always concerned aggregates: individual expectations may err, but errors in different directions roughly cancel. Muth (1961, p. 317) was explicit that the hypothesis does not state "that predictions of entrepreneurs are perfect or that their expectations are all the same." Delegation to a common AI changes this. When every firm forecasts with the same model and the same data, expectations become rational and identical at once. I study what this literal version of rational expectations does to aggregate stability and welfare.

I use a deliberately simple linear-quadratic economy in the tradition of Morris and Shin (2002) and Angeletos and Pavan (2007). Firms choose output before observing demand, and their decisions are strategic complements. A firm either relies on its managers, who observe a private signal, or delegates to an AI that observes a more precise signal common to all users. Delegation replaces the manager's information rather than adding to it. I consider two assumptions about the AI's precision: it is *fixed*, or it is *borrowed*, in the sense that the AI learns from the independent judgment of firms that do not delegate, so its precision falls as adoption rises. Four results follow.

1. **Representative-agent aggregation** (Proposition 1). Dispersed information attenuates the response of aggregate output to fundamentals, more so the stronger the complementarity. Under full delegation, aggregate output equals the AI's conditional expectation of fundamentals for any degree of complementarity. Delegation therefore raises both *exposure*, as idiosyncratic errors are replaced by a common one, and *propagation*, as the attenuation disappears.

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2. **No delegation externality under fixed precision** (Proposition 2). The private and social values of delegating coincide at every adoption share. The additional volatility is the efficient counterpart of better information.

3. **Over-delegation under borrowed precision** (Proposition 3). Each firm that delegates withdraws its private information from what the AI learns, at no cost to itself. This is the logic of Grossman and Stiglitz (1980) applied to an AI that aggregates private information.

4. **A coordination trap** (Proposition 4). Without complementarity, firms stop delegating once the AI is no better informed than a manager. With sufficient complementarity, full delegation remains an equilibrium even when the AI is less informed than any manager, because a firm that deviates is out of step with everyone else. In my baseline example, the market delegates every decision and captures about one-third of the attainable welfare gain.

**Related literature.** The paper builds on work on the social value of public and private information (Morris and Shin, 2002; Angeletos and Pavan, 2007), coordination motives in information acquisition (Hellwig and Veldkamp, 2009), and the attenuation of general-equilibrium effects under dispersed information (Angeletos and Lian, 2018). It applies the view of information as a public good (Hayek, 1945; Grossman and Stiglitz, 1980) to AI. Kleinberg and Raghavan (2021) show that a shared algorithm can lower decision quality even when it outperforms each individual; Danielsson and Uthemann (2025) and Ross et al. (2026) study how AI can amplify financial fragility; and Shumailov et al. (2024) document the degradation of models trained on model-generated data. My contribution is to embed delegation to a common AI, whose precision depends on the private information it displaces, in an economy with complementarity, and to separate the efficient and inefficient parts of the volatility that delegation creates.

## 2 Model

**Payoffs.** Let $\theta \sim N(0, \tau_0^{-1})$ denote efficient output, in deviations. A continuum of firms $i \in [0,1]$ choose output $y_i$, with aggregate $Y = \int y_i \, di$. Firm $i$'s loss is

$$\ell_i = (1-r)(y_i - \theta)^2 + r(y_i - Y)^2, \qquad r \in [0,1), \tag{1}$$

so its best response is $y_i = \mathbb{E}_i[(1-r)\theta + rY]$, and $r$ indexes strategic complementarity, for example through aggregate demand. Welfare is $-\mathbb{E}[\mathcal{L}]$, where

$$\mathcal{L} = \int \ell_i \, di = (1-r)(Y-\theta)^2 + \int (y_i - Y)^2 \, di \tag{2}$$

combines aggregate misallocation and cross-sectional dispersion. This is the efficient benchmark of Angeletos and Pavan (2007): the private and social values of coordination coincide, so the equilibrium use of information is efficient and any inefficiency must come from which information firms act on.

**Information and delegation.** A manager observes $x_i = \theta + \varepsilon_i$, with $\varepsilon_i \sim N(0, \tau_x^{-1})$ independent across firms. The AI observes $s = \theta + \eta$, with $\eta \sim N(0, \tau_s^{-1})$, and every firm that delegates acts on $s$. Before signals are realized, each firm decides whether to delegate, and a fraction $a$ does. Delegation replaces the manager's signal, and all firms know $a$.

**AI precision.** Under *fixed* precision, $\tau_s$ is a constant. Under *borrowed* precision, $\tau_s(a) = \tau_d + \phi(1-a)$ with $\tau_d, \phi > 0$: the AI combines its own data, $\tau_d$, with what it learns from the independent judgment of firms that do not delegate. This is a reduced form for an AI trained on the decisions of non-delegators. Firms take $\tau_s$ as given.

**Lemma 1.** *For given $a$, the unique linear equilibrium has $y = \alpha x_i$ for manager-led firms and $y = \beta s$ for AI-led firms, where $k_x = \tau_x/(\tau_0 + \tau_x)$, $k_s = \tau_s/(\tau_0 + \tau_s)$,*

$$\beta = \frac{(1-r)k_s}{\Delta}, \qquad \alpha = \frac{(1-r)k_x(1 - ra + ra\,k_s)}{\Delta},$$

*with $\Delta = (1 - ra)\big(1 - r(1-a)k_x\big) - r^2 a(1-a)k_x k_s > 0$.*

The proof substitutes $Y = (1-a)\alpha\theta + a\beta s$ into the best responses and matches coefficients. Let $\ell_H(a)$ and $\ell_A(a)$ be the expected losses of a manager-led and an AI-led firm, $G(a) = \ell_H(a) - \ell_A(a)$ the private gain from delegating, and $L(a) = \mathbb{E}[\mathcal{L}] = (1-a)\ell_H(a) + a\ell_A(a)$. A share $a$ is an equilibrium if no firm wants to switch: $G(a) = 0$ for interior $a$, $G(1) \geq 0$ at $a = 1$, or $G(0) \leq 0$ at $a = 0$. A planner chooses $a$ to minimize $L(a)$.

## 3 Results

**Proposition 1** (Aggregation)**.** *(a) Without delegation, $Y = \alpha\theta$ with $\alpha = (1-r)k_x/(1-rk_x)$, which is below the individual Bayesian weight $k_x$ and decreasing in $r$. (b) Under full delegation, $Y = k_s s = \mathbb{E}[\theta \mid s]$ for every $r$.*

*Proof.* Set $a = 0$ and $a = 1$ in Lemma 1. For part (a), $\partial\alpha/\partial r = k_x(k_x - 1)/(1 - rk_x)^2 < 0$, and $\alpha < k_x$ because $k_x < 1$. □

Under dispersed information, a firm is uncertain about what others observe, and complementarity makes that uncertainty costly, so firms respond cautiously and the aggregate under-reacts. Full delegation makes the AI's forecast common knowledge, and the economy aggregates like a representative agent endowed with $s$. Exposure rises, because idiosyncratic errors that wash out in $Y$ are replaced by the common error $\eta$. Propagation rises, because without attenuation $\eta$ enters $Y$ with weight $k_s$. In this sense delegation makes the representative agent of textbook models an accurate description of aggregate behavior, including its exposure to a common forecast error.

**Proposition 2** (No delegation externality under fixed precision)**.** *If $\tau_s$ does not depend on $a$, then $L'(a) = \ell_A(a) - \ell_H(a) = -G(a)$ for all $a$. The private and social values of delegation coincide, and every equilibrium satisfies the planner's first-order conditions.*

*Proof.* (i) For given $a$, the equilibrium strategies minimize $\mathbb{E}[\mathcal{L}]$ among strategies measurable with respect to each firm's information: the first-order condition of this strictly convex team problem is $y_i = \mathbb{E}_i[(1-r)\theta + rY]$. (ii) By the envelope theorem, $L'(a)$ equals the derivative holding $(\alpha, \beta)$ fixed. (iii) Holding strategies fixed, moving firms from managers to the AI changes their losses at rate $\ell_A - \ell_H$ and affects other firms only through $Y$. Since $\partial_Y \int r(y_i - Y)^2 di = -2r \int (y_i - Y)\, di = 0$, this indirect effect is zero. □

Higher volatility under delegation is therefore not evidence of excessive delegation. In every parameterization I have computed, $G$ is increasing in $a$ when $r > 0$, so adoption is self-reinforcing (as in Hellwig and Veldkamp, 2009) and $L$ is concave. The planner's optimum is then the best of the corner equilibria. With multiple equilibria the market can still coordinate on the worse one, but there is no wedge between private and social values at the margin.

**Proposition 3** (Over-delegation under borrowed precision)**.** *If $\tau_s'(a) < 0$, then for every $a \in (0, 1]$,*

$$L'(a) = -G(a) + \frac{\partial L}{\partial \tau_s}\,\tau_s'(a) \;>\; -G(a).$$

*Proof.* The first term follows from Proposition 2. By part (i) of its proof, $L$ is a minimized team loss. A more precise $s$ cannot raise it, because the team could garble $s$, and it strictly lowers it when $a > 0$ because $\beta > 0$. Hence $\partial L/\partial \tau_s < 0$, and the second term is positive. □

The AI is more precise than any manager because it aggregates the judgment of many. Each delegator withdraws its judgment from that aggregate without bearing the cost, so the marginal private value of delegation exceeds its social value at every adoption share.

**Proposition 4** (Coordination trap)**.** *Let precision be borrowed with $\tau_d < \tau_x$, so that under full delegation the AI is less informed than any manager. (a) If $r = 0$, full delegation is not an equilibrium: firms delegate only while $\tau_s(a) \geq \tau_x$. (b) If $r$ is sufficiently close to one, full delegation is an equilibrium.*

*Proof.* (a) With $r = 0$, $\ell_H = 1/(\tau_0 + \tau_x)$ and $\ell_A = 1/(\tau_0 + \tau_s)$, so $G(a) \geq 0$ if and only if $\tau_s(a) \geq \tau_x$. (b) At $a = 1$, $\ell_A = (1-r)/(\tau_0 + \tau_d) \to 0$ as $r \to 1$. A deviating manager-led firm chooses $y = c\, x_i$ with $c = [(1-r) + rk_s]k_x$, and $\ell_H \geq r\,\mathbb{E}(cx_i - k_s s)^2 \to k_s^2\left[(1-k_x)^2/\tau_0 + k_x^2/\tau_x + 1/\tau_d\right] > 0$. Hence $G(1) > 0$ for $r$ close to one. □

# 4 An Illustration

The baseline sets $\tau_0 = \tau_x = 1$ and $r = 0.4$. Under borrowed precision, $\tau_d = 0.75$ and $\phi = 4$, so the AI starts at $\tau_s = 4.75$ in both cases. These values are illustrative rather than calibrated. In this example the trap of Proposition 4 binds for any $r > 0.13$.

**Table 1:** Aggregate outcomes in the baseline

| | | Fixed precision | Borrowed precision | |
|---|---|---|---|---|
| | No AI | Market = planner | Market | Planner |
| Share of decisions delegated, $a$ | 0 | 1 | 1 | 0.72 |
| AI precision relative to a manager, $\tau_s/\tau_x$ | — | 4.75 | 0.75 | 1.88 |
| Response of $Y$ to $\theta$ | 0.38 | 0.83 | 0.43 | 0.54 |
| Standard deviation of $Y$ (no AI = 100) | 100 | 242 | 175 | 166 |
| Share of $\mathrm{Var}(Y)$ due to the AI's error | 0% | 17% | 57% | 25% |
| RMS distance of $Y$ from $\theta$ (no AI = 100) | 100 | 67 | 121 | 89 |
| Reduction in welfare loss relative to no AI | 0% | 72% | 9% | 26% |

With fixed precision, full delegation raises the standard deviation of output by a factor of 2.4, yet output tracks fundamentals more closely and the welfare loss falls by 72 percent: volatility rises while waste falls (Table 1). With borrowed precision, the private gain from delegating stays positive at every adoption share (Figure 1a), including beyond 94 percent, where the AI becomes less precise than a manager. The market delegates fully. In that outcome, 57 percent of the variance of output reflects the AI's error, output strays 21 percent further from the efficient level than without AI, and the market captures one-third of the attainable gain (Figure 1b). The planner delegates 72 percent and keeps the AI at 1.88 times a manager's precision. If $\tau_d < 0.6$, full delegation is worse than having no AI. Across $r \in \{0.2, 0.4, 0.6\}$ and $\tau_d \in \{0.5, 0.75, 1, 1.5\}$, the market delegates at least 97 percent of decisions while the planner delegates between 60 and 100 percent. The shortfall is largest when $\tau_d$ is low and shrinks as $r$ rises, because stronger complementarity also raises the social value of coordination.

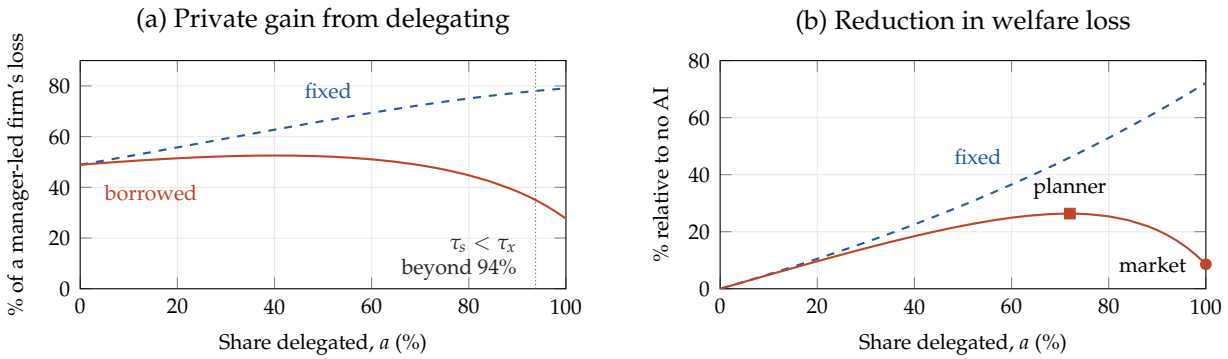



**Figure 1:** Baseline. Dashed lines: fixed precision. Solid lines: borrowed precision. Under borrowed precision the gain from delegating stays positive even after the AI becomes less precise than a manager, and the market ($a = 1$) captures one-third of the gain available to the planner ($a = 0.72$).

## 5 Policy Implications and Next Steps

The model treats independent judgment as a public input to AI, and that is where policy should aim. In the baseline, a floor that keeps 28 percent of decisions independent attains the planner's allocation. A Pigouvian tax on delegation is less reliable. The tax that supports $a = 0.72$ exceeds the private gain to the first adopter, so no adoption becomes an equilibrium as well. Policies aimed at AI accuracy or at the number of providers miss the mechanism.

The model is static, the AI's learning is in reduced form, and there is a single AI. Three extensions are natural: micro-founding $\tau_s(a)$ by letting the AI learn from the cross-section of non-delegators' actions; a dynamic version in which adoption and learning co-evolve, where the trap may arrive after a period of apparent stability; and evidence from controlled markets populated by AI agents, which can vary the adoption share and the information agents share directly.

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